

TITAN SMALL BUSINESS SERVICES: SECURITY ARCHITECTURE DOCUMENT (SAD)

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1. Perimeter Hardening (UFW & SSH)

- **SSH Status:** Disabled root login and password-based authentication in `/etc/ssh/sshd_config`. Enforced Public Key Authentication to prevent brute-force attacks.
- **Firewall Logic:** Implemented a 'Default Deny' incoming policy using UFW. Explicitly allowed Port 22 (SSH) for management and Port 8080 (Web) for application traffic.

2. The Automated Auditor (Python)

- **Script Logic:**

```
import os
disk_status = os.popen("df -h").read()
hostname = "10.0.2.15"
response = os.system(f"ping -c 4 {hostname}")
with open("/var/log/dc_audit.log", "a") as log:
    log.write("\n--- SYSTEM AUDIT ---\n")
    log.write("DISK USAGE:\n" + disk_status)
    if response == 0:
        log.write("NETWORK STATUS: DC is UP\n")
    else:
        log.write("NETWORK STATUS: DC is DOWN\n")
```

- **Telemetry Path:** `/var/log/dc_audit.log`

3. Containerized App (Docker)

- **Network Isolation:** I deployed a dual-network bridge architecture. The 'wiki' service is connected to both frontend and backend networks, while the 'db' service is isolated strictly on the internal 'backend' network with no host port mapping, effectively air-gapping the data layer from external access.
- **Stack Health:** NAME IMAGE STATUS PORTS wiki_frontend nginx Up 30 minutes 0.0.0.0:8080->80/tcp db_backend mysql Up 30 minutes (No ports exposed)

4. Executive Summary

The Hardened Outpost utilizes a defense-in-depth strategy to secure Titan SBS infrastructure against unauthorized access. By combining host-level firewall

restrictions, containerized network segmentation, and automated health auditing, the environment maintains a high level of integrity. The current posture ensures that sensitive backend resources are unreachable from the public internet while providing reliable service delivery via the hardened frontend.